

Отчёт по лабораторной работе №4

Продвинутое использование git

Сарханов Рамиэль Маркс оглы

Содержание

1	Цель работы	5
2	Теоретические сведения	6
3	Выполнение лабораторной работы	8
3.1	Работа с тестовым репозиторием	8
3.2	Подготовка рабочего репозитория	19
4	Вывод	21

Список иллюстраций

3.1	Node.js	9
3.2	установка commitizen	10
3.3	установка standard-changelog	11
3.4	package.json	12
3.5	Отправка	13
3.6	Инициализация и загрузка изменений	14
3.7	Завершение релиза	15
3.8	Отправка	16
3.9	Объединение веток	17
3.10	Завершение релиза	18
3.11	package.json и коммит	19
3.12	Завершение релиза	20

Список таблиц

1 Цель работы

Получение навыков правильной работы с репозиториями git.

2 Теоретические сведения

- Gitflow Workflow опубликована и популяризована Винсентом Дриссеном.
- Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта.
- Данная модель отлично подходит для организации рабочего процесса на основе релизов.
- Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.
- Последовательность действий при работе по модели Gitflow:
- Из ветки master создаётся ветка develop.
- Из ветки develop создаётся ветка release.
- Из ветки develop создаются ветки feature.
- Когда работа над веткой feature завершена, она сливается с веткой develop.
- Когда работа над веткой релиза release завершена, она сливается в ветки develop и master.
- Если в master обнаружена проблема, из master создаётся ветка hotfix.

- Когда работа над веткой исправления hotfix завершена, она сливается в ветки develop и master.

3 Выполнение лабораторной работы

3.1 Работа с тестовым репозиторием

Для работы с Node.js добавим каталог с исполняемыми файлами, устанавливаемыми yarn, в переменную PATH.


```
ramielsarhanov@ramielsarhanov:~$ pnpm setup
Appended new lines to /home/ramielsarhanov/.bashrc

Next configuration changes were made:
export Pnpm_HOME="/home/ramielsarhanov/.local/share/pnpm"
case "$PATH:" in
  *"$Pnpm_HOME:") ;;
  *) export PATH="$Pnpm_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/ramielsarhanov/.bashrc
ramielsarhanov@ramielsarhanov:~$ source ~/.bashrc
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 3.1: Node.js

Программа commitizen используется для помощи в форматировании коммитов. При этом устанавливается скрипт git-cz, который мы и будем использовать для коммитов.

```
ramielsarhanov@ramielsarhanov:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/ramielsarhanov/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
Done in 2.7s using pnpm v10.27.0
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 3.2: установка commitizen

Программа standard-changelog используется для помощи в создании логов.

```
ramielsarhanov@ramielsarhanov:~$ pnpm add -g standard-changelog
WARN GET https://registry.npmjs.org/conventional-changelog error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29223ms: https://registry.npmjs.org/conventional-changelog
WARN GET https://registry.npmjs.org/@simple-lib%2Fstream-utils error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29222ms: https://registry.npmjs.org/@simple-lib%2Fstream-utils
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
+++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/ramielsarhanov/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 59.1s using pnpm v10.27.0
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 3.3: установка standard-changelog

Делаем первый коммит и выкладываем на github.

Необходимо заполнить несколько параметров пакета.

Таким образом, файл package.json приобретает вид:



```
1 {  
2   "name": "git-extended",  
3   "version": "1.0.0",  
4   "description": "Git repo for educational purposes",  
5   "main": "index.js",  
6   "repository": "git@github.com:ramiel-sarhanov/git-extended.git",  
7   "author": "ramiel-sarhanov",  
8   "license": "CC-BY-4.0",  
9   "config": {  
10     "commitizen": {  
11       "path": "cz-conventional-changelog"  
12     }  
13   }  
14 }
```

Рисунок 3.4: package.json

Добавим новые файлы.

Выполним коммит.

Отправим на github.

```

ramielsarhanov@ramielsarhanov:~/git-extended$ git add .
ramielsarhanov@ramielsarhanov:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:  Other changes that don't modify
src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master cf40bd4] chore: test
1 file changed, 14 insertions(+)
create mode 100644 package.json
ramielsarhanov@ramielsarhanov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
19ea55f..cf40bd4 master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$

```

Рисунок 3.5: Отправка

Инициализируем git-flow

Проверьте, что Вы на ветке develop

Загрузите весь репозиторий в хранилище

```

ramielsarhanov@ramielsarhanov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/ramielsarhanov/git-extended/.git/hooks]
ramielsarhanov@ramielsarhanov:~/git-extended$ git branch
* develop
  master
ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/ramiel-sarhanov/git-extended/pull/new/develop
remote:
To github.com:ramiel-sarhanov/git-extended.git
 * [new branch]      develop -> develop
ramielsarhanov@ramielsarhanov:~/git-extended$

```

Рисунок 3.6: Инициализация и загрузка изменений

Установите внешнюю ветку как вышестоящую для этой ветки

Создадим релиз с версией 1.0.0

Создадим журнал изменений

Добавим журнал изменений в индекс

Зальём релизную ветку в основную ветку

```

ramielsarhanov@ramielsarhanov:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

ramielsarhanov@ramielsarhanov:~/git-extended$ standard-changelog --first-release
ramielsarhanov@ramielsarhanov:~/git-extended$ git add CHANGELOG.md
ramielsarhanov@ramielsarhanov:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 b201f78] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
Already on 'master'

```

Рисунок 3.7: Завершение релиза

Отправим данные на github

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.68 KiB | 2.68 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
   cf40bd4..eb20261  develop -> develop
   cf40bd4..c00aeba  master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
 * [new tag]         1.0.0 -> 1.0.0
ramielsarhanov@ramielsarhanov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/ramiel-sarhanov/git-extended/releases/tag/v1.0.0
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 3.8: Отправка

Создадим ветку для новой функциональности По окончании разработки новой функциональности следующим шагом следует объединить ветку `feature_branch` с `develop`:


```

ramielsarhanov@ramielsarhanov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

ramielsarhanov@ramielsarhanov:~/git-extended$ touch test2
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was eb20261).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/git-extended$

```

Рисунок 3.9: Объединение веток

Создадим релиз с версией 1.2.3

Обновите номер версии в файле package.json. Установите её в 1.2.3

Создадим журнал изменений

Добавим журнал изменений в индекс

Зальём релизную ветку в основную ветку

```
> Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.79 KiB | 2.79 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/git-extended.git
   eb20261..345bbe5  develop -> develop
   c80aeba..a2fec6f  master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 3.10: Завершение релиза

3.2 Подготовка рабочего репозитория

```
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git add .
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore: Other changes that don't modify
src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master dbd347b] chore: test
1 file changed, 11 insertions(+), 11 deletions(-)
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
b5ecd08..dbd347b master -> master
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$
```

Рисунок 3.11: package.json и коммит

```

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.77 KiB | 2.77 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
   dbd347b..c0e8d19  develop -> develop
   dbd347b..36242d6  master -> master
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
 * [new tag]         1.0.0 -> 1.0.0
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$

```

Рисунок 3.12: Завершение релиза

4 Вывод

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.